

Hello Ben, I got some new understanding about the lumibot project that I want to share with you.

This document discuss some understanding of lumibot workflow. As shown in Figure 1

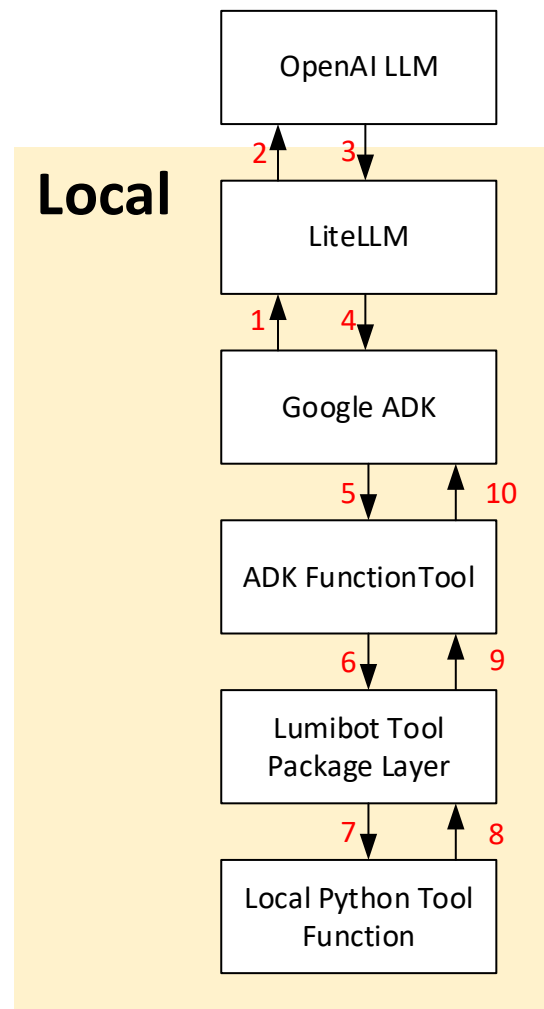


Figure 1 definition of the each modules

There are 6 modules and 10 arrows. Following are some understanding for each module:

1. OpenAI LLM (do not have to be OpenAI)
OpenAI LLM is the “brain”, who is able to think. It read prompt, tool description, decide use tools, and write summary
2. LiteLLM
LiteLLM is the API adapter. On one hand, it converts the info from Google ADK to the format that OpenAI LLM can take as input; On the other hand, it converts the info from OpenAI LLM to the format that Google ADK can take as input.
3. Google ADK
Google ADK is the agent frame. It organise prompt, context, tool definition, multiple rounds discussion history.

It is also the answer that you ask me that where the context stored. The context is stored here, but it is only in RAM, and it is deleted after the agent run.

4. ADK Function Tool

ADK Function Tool is the Tool Execution Entrance. It receives the tool list and parameters that LLM want to use, find the tools, do the parameter format conversion.

5. Lumibot Tool Package Layer

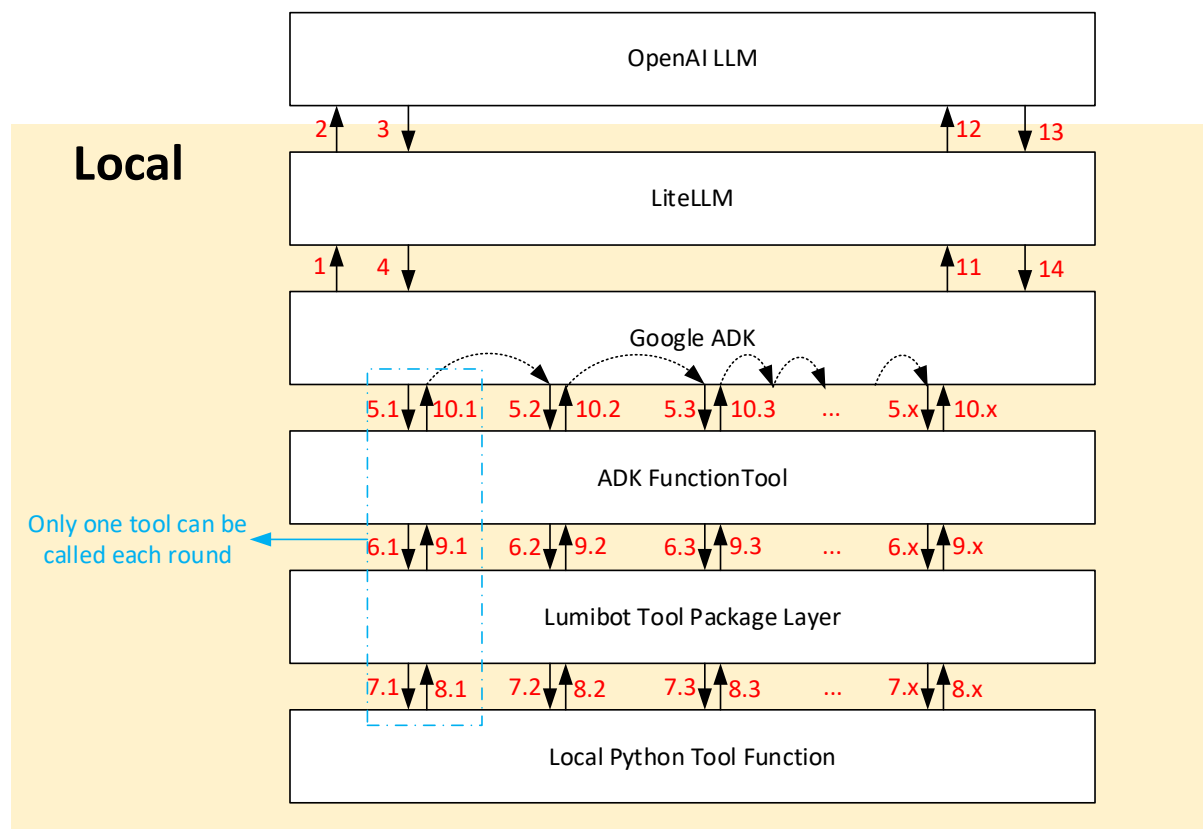
Lumibot Tool Package Layer plays an adapter that converts the command from ADK Function Tool to local executable command to local tools.

Basically, I think both ADK Function Tool and Lumibot Tool Package Layer are playing adapter role. ADK Function Tool is at the side of Google ADK, and Lumibot Tool Package Layer is at Lumibot side

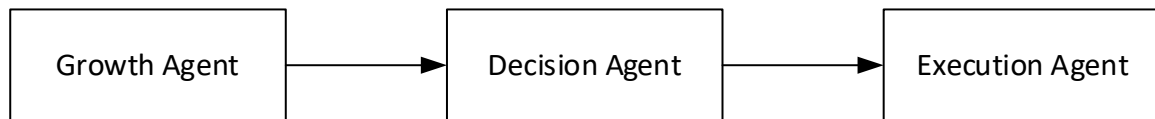
6. Local Python Tool Function

Local Python Tool Function is executing the local Python functions, for example, check the account info, search news, load history data, making orders, check orders, etc.

In practice, one round can call multiple tools, but the arrows 5, 6, 7, 8, 9, and 10 can only call the tool one by one. So that the actual workflow is as Figure 2.



What I am going to do next is use simplified agent work flow:



Run a back test, and take it as an example, learn what info is in each arrow.

Please let me know if any questions or do you think are we on the right track.

Thanks

Ran